Data Structures Handout: Stacks and Queues

Course IT46-105 · Week 5 · All lab code in this course uses Python 3.12

1. Stack

A stack is a last-in, first-out (LIFO) collection. push, pop and peek all run in O(1) time. Typical uses are undo histories, the function call stack and evaluating postfix expressions. Example: the postfix expression "5 3 + 2 *" evaluates to 16.

2. Queue

A queue is a first-in, first-out (FIFO) collection. With a circular buffer, enqueue and dequeue run in O(1). Breadth-first search and print spooling use queues.

In our lab the circular queue has a capacity of 8 slots. One slot is always left empty to tell a full queue from an empty one, so it holds at most 7 items. The queue is full when (rear + 1) % capacity == front.

3. Priority queue and deque

A priority queue built on a binary heap inserts in O(log n) and removes the minimum in O(log n). A deque (double-ended queue) allows insertion and removal at both ends.

4. Lab 5

Implement a stack using two queues. Submit a single file named stack_queue.py before the end of week 6. Submissions are checked by an autograder that runs 12 hidden test cases.